

School of Engineering and Applied Science (SEAS), Ahmedabad University
CSE 400: Fundamentals of Probability in Computing
Project Scribe Submission
Group Number: S1 G7 ITS
Milestone 2

Scribe Question 1: Project System and Objective

Answer:

In Milestone 2, our project focuses on formally modeling the probabilistic behavior of a signalized four-approach Indian intersection. The primary objective of the system remains:

$$\min E[\bar{W}]$$

where

$$\bar{W} = \frac{1}{4} \sum_{i=1}^4 W_i$$

and W_i denotes the waiting time per lane.

The system dynamically adjusts green light durations using a Las Vegas randomized extension mechanism layered over a Webster-based cycle computation.

The primary sources of uncertainty are:

- Random vehicle arrivals
- Variability in queue evolution
- Randomized green-time extension
- Variation in vehicle speeds
- Variability in effective saturation flow

Distribution of Queue Length

Let Q_i denote the steady-state queue length at lane i .

Under Poisson arrivals and M/M/1 steady-state approximation, traffic intensity is:

$$\rho_i = \frac{\lambda_i}{\mu_i}, \quad 0 < \rho_i < 1$$

Probability Mass Function (PMF):

$$P(Q_i = n) = (1 - \rho_i)\rho_i^n, \quad n = 0, 1, 2, \dots$$

This shows that queue length follows a geometric distribution.

Cumulative Distribution Function (CDF):

$$F(n) = P(Q_i \leq n)$$

$$F(n) = \sum_{k=0}^n (1 - \rho_i) \rho_i^k$$

Using geometric series:

$$F(n) = 1 - \rho_i^{n+1}$$

Expectation and Variance:

$$E[Q_i] = \frac{\rho_i}{1 - \rho_i}$$

$$\text{Var}(Q_i) = \frac{\rho_i}{(1 - \rho_i)^2}$$

Inference:

- As $\rho_i \rightarrow 1$, expected queue length grows rapidly.
- Tail probability $P(Q_i > n) = \rho_i^{n+1}$ increases sharply with traffic intensity.
- Small increases in arrival rate can significantly increase congestion risk.

Thus, the geometric distribution of queue length provides a probabilistic characterization of congestion escalation.

Scribe Question 2: Key Random Variables and Uncertainty Modeling

Answer:

The key random variables in our system are:

1. Number of Vehicles $N_i(t)$

Vehicle arrivals at lane i in time interval t follow a Poisson distribution:

$$N_i(t) \sim \text{Poisson}(\lambda_i t)$$

PMF:

$$P(N_i = k) = \frac{(\lambda_i t)^k e^{-\lambda_i t}}{k!}$$

2. Queue Length $Q_i(t)$

Queue length evolves as:

$$Q_i(t+1) = \max\{Q_i(t) + N_i(t) - D_i(t), 0\}$$

Assuming traffic stability:

$$\rho_i = \frac{\lambda_i}{\mu_i} < 1$$

the stationary distribution of queue length is geometric:

$$P(Q_i = n) = (1 - \rho_i)\rho_i^n$$

3. Saturation Weight Factor Y_i

Defined as:

$$Y_i = \frac{Q_i}{S_i}$$

where S_i is saturation flow. Since Q_i is geometric, Y_i is a scaled geometric random variable.

4. Waiting Time per Lane W_i

Using Little's Law:

$$E[W_i] = \frac{E[Q_i]}{\lambda_i}$$

Under M/M/1 approximation:

$$W_i \sim \text{Exponential}(\mu_i - \lambda_i)$$

PDF:

$$f_{W_i}(w) = (\mu_i - \lambda_i)e^{-(\mu_i - \lambda_i)w}$$

5. Vehicle Speed V_i

Vehicle speed variability is modeled as:

$$V_i \sim \mathcal{N}(\mu_v, \sigma_v^2)$$

reflecting heterogeneous Indian traffic behavior.

6. Green-Time Extension ΔG

$$\Delta G \sim U\{0, 1, 2, 3, 4, 5\}$$

PMF:

$$P(\Delta G = k) = \frac{1}{6}$$

Scribe Question 3: Probabilistic Inequalities and Bounds

Answer:

We now apply probabilistic inequalities to quantify congestion risk.

(1) Jensen's Inequality

Let congestion cost be:

$$C(Q_t) = Q_t^2$$

Since $f(x) = x^2$ is convex:

$$E[Q_t^2] \geq (E[Q_t])^2 \tag{3}$$

Substituting expectation:

$$E[Q_t^2] \geq \left(\frac{\rho}{1-\rho} \right)^2 \tag{4}$$

This inequality shows nonlinear growth of congestion cost.

(2) Markov's Inequality

For any non-negative random variable Q_t and any threshold $a > 0$:

$$P(Q_t \geq a) \leq \frac{E[Q_t]}{a} \tag{1}$$

Substituting expectation:

$$P(Q_t \geq a) \leq \frac{\frac{\rho}{1-\rho}}{a} \tag{2}$$

This provides an upper bound on congestion probability.